

Applications of Determinants

Cramer's Rule: let $A \in \mathbb{R}^{n \times n}$ be such that $|A| \neq 0$.
Then $A\vec{x} = \vec{b}$ is satisfied when

$$x_i = \frac{|A_i|}{|A|}$$

*i*th column

where $A_i = [A_{\cdot,1} \cdots A_{\cdot,i-1} \quad \vec{b} \quad A_{\cdot,i+1} \cdots A_{\cdot,n}]$

Proof: Noting $x_i = |\vec{e}_1 \cdots \vec{e}_{i-1} \quad \vec{x} \quad \vec{e}_{i+1} \cdots \vec{e}_n|$
we have that $A[\vec{e}_1 \cdots \vec{e}_{i-1} \quad \vec{x} \quad \vec{e}_{i+1} \cdots \vec{e}_n] = A_i$
and thus taking determinants we have that
 $|A| \cdot x_i = |A_i|$ and thus $x_i = \frac{|A_i|}{|A|}$.

Exercise

1) Use Cramer's rule to solve the system

$$\begin{aligned} 3x_1 + x_3 &= -2 \\ 2x_1 - x_2 + 4x_3 &= 11 \\ x_2 - 2x_3 &= -7 \end{aligned}$$

$$\vec{x} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$$

$$\text{As } A = \begin{bmatrix} 3 & 0 & 1 \\ 2 & -1 & 4 \\ 0 & 1 & -2 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} -2 \\ 11 \\ -7 \end{bmatrix}$$

we have that

$$\begin{aligned} x_1 &= \frac{\begin{vmatrix} -2 & 0 & 1 \\ 11 & -1 & 4 \\ -7 & 1 & -2 \end{vmatrix}}{\begin{vmatrix} 3 & 0 & 1 \\ 2 & -1 & 4 \\ 0 & 1 & -2 \end{vmatrix}} = \frac{-2 \begin{vmatrix} -1 & 4 \\ 1 & -2 \end{vmatrix} + \begin{vmatrix} 11 & -1 \\ -7 & 1 \end{vmatrix}}{3 \begin{vmatrix} -1 & 4 \\ 1 & -2 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix}} \\ &= \frac{4 + 4}{-6 + 2} = -2 \end{aligned}$$

and so on

Adjugate Matrix

Given $A \in \mathbb{R}^{n \times n}$ let $\text{adj}(A) \in \mathbb{R}^{n \times n}$ be such that

$$(\text{adj}(A))_{ij} := (-1)^{i+j} |M_{ji}|$$

(remove row j and column i)

$\text{adj}(A)^T$ is the cofactor matrix

Theorem 13.1 If A is invertible, i.e. $|A| \neq 0$,

$$\text{then } A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

$$(\text{adj}(A) \cdot A = |A| \cdot I)$$

$$\text{Proof: Since } (A \cdot \text{adj}(A))_{i,j} = \sum_{k=1}^n (-1)^{j+k} A_{i,k} |M_{j,k}|$$

$$\text{we have } (A \cdot \text{adj} A)_{i,j} = \begin{cases} |A| & i=j \\ 0 & i \neq j \end{cases}$$

$$\text{as } \sum_{k=1}^n (-1)^{j+k} A_{i,k} |M_{j,k}| \text{ is the determinant}$$

of the matrix where the j th row of A

is replaced by the i th row of A .

However if A is not invertible so that $|A|=0$
we have that $A \cdot \text{adj}(A)$ is the zero matrix
and hence $\text{range}(\text{adj} A) \subseteq \text{null } A$.

Similarly for A^T , as $(A \cdot \text{adj} A)^T = A^T (\text{adj} A)^T$
is the zero matrix we have that

$$\text{range}(\text{adj } A)^T \subseteq \text{null } A^T$$

Exercise

2) Given $A = \begin{bmatrix} 2 & -1 & 4 \\ 1 & 3 & -2 \\ 3 & 2 & 2 \end{bmatrix}$ Compute $\text{adj}(A)$
and use it to find $\text{null } A$ and $\text{null } A^T$

$$|M_{11}| = \begin{vmatrix} 3 & -2 \\ 2 & 2 \end{vmatrix} = 10 \quad |M_{21}| = \begin{vmatrix} -1 & 4 \\ 2 & 2 \end{vmatrix} = -10 \quad |M_{31}| = \begin{vmatrix} -1 & 4 \\ 3 & -2 \end{vmatrix} = -10$$

$$|M_{12}| = \begin{vmatrix} 1 & -2 \\ 3 & 2 \end{vmatrix} = 8 \quad |M_{22}| = \begin{vmatrix} 2 & 4 \\ 3 & 2 \end{vmatrix} = -8 \quad |M_{32}| = \begin{vmatrix} 2 & 4 \\ 1 & -2 \end{vmatrix} = -8$$

$$|M_{13}| = \begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} = -7 \quad |M_{23}| = \begin{vmatrix} 2 & -1 \\ 3 & 2 \end{vmatrix} = 7 \quad |M_{33}| = \begin{vmatrix} 2 & -1 \\ 1 & 3 \end{vmatrix} = 7$$

$$\text{Now } \text{adj}(A) = \begin{bmatrix} 10 & 10 & -10 \\ -8 & -8 & 8 \\ -7 & -7 & 7 \end{bmatrix} \quad \text{and thus}$$

$$\text{null } A = \text{range}(\text{adj } A) = \text{span} \left(\begin{bmatrix} 10 \\ -8 \\ -7 \end{bmatrix} \right)$$

$$\text{and } \text{null } A^T = \text{range}(\text{adj } A)^T = \text{span} \left(\begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} \right)$$

Intro to Eigenvalues and Eigenvectors

Invariant subspace

Def 13.2 Let $T \in \mathcal{L}(V, V)$. Then a subspace W of V is a T -invariant subspace if $T(W) \subseteq W$.

Example: For any V -endomorphism T ($T \in \mathcal{L}(V, V)$) $\text{range}(T)$ and $\text{null}(T)$ are T -invariant subspaces.

1-dim invariant subspace: let $W = \text{span}\{\vec{v}\}$ where $\vec{v} \neq \vec{0}$. Then W is T -invariant if $T\vec{v} = \lambda\vec{v}$. If so we have that

$$\vec{0} = T\vec{v} - \lambda\vec{v} = (T - \lambda I)\vec{v}$$

and thus $\vec{v} \in \ker(T - \lambda I)$

$$(T - \lambda I)(\vec{x}) = T\vec{x} - \lambda\vec{x}$$

Def 13.3 A non zero vector $\vec{v}$ is an e-vector of T with e-value of λ iff $\vec{v} \in \ker(T - \lambda I)$

Theorem 13.4 Let $T \in \mathcal{L}(\mathbb{C}^n, \mathbb{C}^n)$ and let $A = [T]_{\mathcal{B}}$. Then λ is an e-value of A iff $|A - \lambda I| = 0$ (λ are independent $\mathcal{B}$)

Def 13.5 The polynomial $p(\lambda) = p_A(\lambda) = |A - \lambda I|$ is the characteristic polynomial of A and has degree equal to the size of A (# rows)

Now λ_0 is an e-value of A iff $p_A(\lambda_0) = 0$

Def 13.6 If λ_0 is an e-value of A then $\text{null}(A - \lambda_0 I)$ is the eigenspace of A corresponding to λ_0 .

If $\{\vec{v}_{0_1}, \dots, \vec{v}_{0_k}\}$ is a basis for $\text{null}(A - \lambda_0 I)$ then we say that $\vec{v}_{0_1}, \dots, \vec{v}_{0_k}$ are the* e-vectors of A associated to λ_0 . * e-vectors are not unique

Eigenvalues of Special Matrices

- i. Identity : I has an e-value of 1 with every vector being an e-vector
- ii. Diagonal, Upper and Lower triangular matrices: The e-values are the diagonal
- iii. Projections: If $P^2 = P$, then P has e-values of 1 and 0.
- iv. Orthogonal matrices: If $A \in \mathbb{R}^{n \times n}$ is orthogonal then $P_A(\lambda) = \lambda^n \pm 1$

Theorem 13.7 E-vectors corresponding to distinct e-values are linearly independent.

Proof: Let $\{\vec{v}_1, \dots, \vec{v}_k\}$ be non zero e-vectors of A such that $A\vec{v}_i = \lambda_i \vec{v}_i$ where $\lambda_i \neq \lambda_j$ for all $i \neq j$. Now assume that $\vec{v}_k = \sum_{i=1}^{k-1} c_i \vec{v}_i$ and that $\{\vec{v}_1, \dots, \vec{v}_{k-1}\}$ are linearly independent. Now as

$$\lambda_k \vec{v}_k = A \left(\sum_{i=1}^{k-1} c_i \vec{v}_i \right)$$

$$= \sum_{i=1}^{k-1} c_i A \vec{v}_i$$

$$= \sum_{i=1}^{k-1} c_i \lambda_i \vec{v}_i$$

and thus

$$\vec{0} = \left(\sum_{i=1}^{k-1} c_i \lambda_i \vec{v}_i \right) - \lambda_k \vec{v}_k$$

$$= \sum_{i=1}^{k-1} c_i \underbrace{(\lambda_i - \lambda_k)}_{=0} \vec{v}_i$$

$\neq 0$

Now $c_i = 0$ for all i and thus $\vec{v}_k = \vec{0}$ which is a contradiction.

Exercise

3) Find the e-values and e-vectors of the following matrices

a. $A = \begin{bmatrix} 3 & -4 \\ 3 & 11 \end{bmatrix}$

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} 3-\lambda & -4 \\ 3 & 11-\lambda \end{vmatrix} = (3-\lambda)(11-\lambda) + 12 \\ &= \lambda^2 - 14\lambda + 45 \\ &= (\lambda-9)(\lambda-5) \end{aligned}$$

$$\lambda = 9, 5$$

$$\lambda_1 = 9: A - 9I = \begin{bmatrix} -6 & -4 \\ 3 & 2 \end{bmatrix}, \vec{v}_1 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

interchange the components and negate one

$$\lambda_2 = 5: A - 5I = \begin{bmatrix} -2 & -4 \\ 3 & 6 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$b. \quad B = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 11 \end{bmatrix}$$